

AG Grid Enterprise vs Custom Prototype Feature Analysis

Executive Summary

This document analyzes how much of the current Forecasting & Planning grid prototype can be implemented using AG Grid Enterprise **without customizations**, versus what requires custom development.

~40-45%

can be implemented out-of-the-box with AG Grid Enterprise

~55-60%

requires significant custom development

Features That AG Grid Enterprise Supports Out-of-the-Box

Feature	AG Grid Support	Implementation	Notes
Hierarchical Data Display	✓ 95%	Tree Data mode with <code>getDataPath()</code>	AG Grid handles expand/collapse, indentation, icons
Cell Editing	✓ 90%	Set <code>editable: true</code> on columns	AG Grid handles input, validation, save/cancel
Frozen Columns	✓ 100%	<code>pinned: 'left'</code> on first column	Perfect match - built-in feature
Sticky Header Row	✓ 100%	Default behavior	Perfect match - no config needed
Column Resizing	✓ 100%	<code>resizable: true</code> on columns	Better than slider implementation
Row Selection	✓ 100%	<code>rowSelection: 'single'</code>	Perfect match
Context Menu	✓ 90%	<code>getContextMenuItems()</code> (Enterprise)	AG Grid handles positioning, styling
Column Filtering	✓ 85%	<code>filter: true</code> or custom filters	Rich filter UI available
Column Sorting	✓ 100%	<code>sortable: true</code> on columns	Perfect match
Cell Styling	✓ 80%	<code>cellStyle</code> and <code>cellClass</code> functions	Can style backgrounds, borders, text colors
Row Styling	✓ 85%	<code>rowStyle</code> and <code>rowClass</code> functions	Can set row backgrounds, hover effects
Cell Renderers	✓ 70%	Custom React components	Requires component development
Search/Filtering	✓ 75%	Quick Filter or custom filter	Search available, highlighting needs custom renderer

Features Requiring Significant Customization

Feature	AG Grid Support	Required Customization	Effort Level
Complex Value Propagation	✗ 0%	Custom <code>onCellValueChanged</code> handler, upward/downward propagation, time aggregation, cross-measure dependencies	High - Core business logic
Cell State Management	⚠ 40%	External state tracking, custom cell renderer, styling based on state	Medium-High - State + renderers
Visual Indicators	⚠ 30%	Custom cell renderer, SVG icons (arrows, locks), CSS positioning (triangles)	Medium - Custom renderer
Delta Badge Display	⚠ 30%	Calculate delta percentage, render badge component, position relative to value	Medium - Custom renderer
Adjustment Notes	✗ 0%	Custom cell editor, dropdown/popover, keyboard navigation, state management	High - Complex editor
Edit History Tracking	✗ 0%	External state management, store notes/timestamps/user info, display in side panel	Medium-High - External state
Cell Details Panel	✗ 0%	Separate React component, integrate with cell selection, display history/discussions	Medium - Separate component
Threaded Discussions	✗ 0%	Separate React component, comment/reply functionality, data storage	Medium - Separate feature
Cell Locking	⚠ 50%	Dynamic <code>editable</code> function, prevent propagation (custom logic), visual lock icon	Medium - Logic + renderer
Multiple Layout Views	⚠ 40%	Transform data structure, reconfigure columns dynamically, maintain state	Medium-High - Data transformation
Column Width Auto-Expansion	✗ 0%	Measure cell content width, dynamically adjust widths, handle resize events	Medium - Custom logic
Search Highlighting	⚠ 30%	Parse search terms, highlight matching text in renderer, handle multiple matches	Medium - Custom renderer
Readonly Texture	⚠ 20%	CSS background pattern, disable editing via <code>editable</code> function	Low-Medium - CSS + function

Feature	AG Grid Support	Required Customization	Effort Level
Cell Info Popover	⚠️ 30%	Custom renderer or overlay, position relative to cell, display edit history	Medium - Custom component
Settings Panel	❌ 0%	Separate React component, update column definitions, handle layout changes	Medium - Separate component

Detailed Feature Breakdown

✅ What AG Grid Enterprise Provides Out-of-the-Box (~40-45%)

- **Grid Infrastructure:** Hierarchical display, editing, scrolling, resizing
- **Basic Interactions:** Selection, context menu, filtering, sorting
- **Styling Capabilities:** Cell/row styling, custom renderers (with development)
- **Enterprise Features:** Advanced filtering, column menu, tool panels

⚠️ What Requires Custom Development (~55-60%)

- **Core Business Logic:** Value propagation, cross-measure dependencies, time aggregation
- **Custom UI Components:** Adjustment notes, edit history panel, threaded discussions
- **State Management:** Cell states (edited/impacted/saved), edit tracking, notes
- **Visual Indicators:** Arrows, lock icons, note triangles, delta badges
- **Advanced Features:** Auto column width, search highlighting, multiple layouts

Conclusion & Recommendation

AG Grid Enterprise can handle ~40-45% of this prototype without customization, primarily:

- Grid structure and layout
- Basic editing and interactions
- Column/row management
- Enterprise filtering and sorting

However, ~55-60% requires significant custom development, including:

- All business logic (propagation, calculations)
- Custom UI components (notes, history, discussions)
- Visual indicators and state management
- Advanced features (auto-width, highlighting)

Key Insight: While AG Grid Enterprise provides a solid foundation, this prototype's unique value lies in its **custom business logic and specialized UI components**, which would need to be built regardless of the grid library chosen.

Recommendation: AG Grid Enterprise would reduce development time for basic grid functionality (~40-45%), but the core differentiating features (value propagation, edit history, notes, visual indicators) would still require the

same amount of custom development effort.

Forecasting & Planning Tool – AG Grid Enterprise Feature Analysis

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